Agent 001

Hard Drives, Soft Prices

"Prices are falling," we say for almost everything, but it's not true for anything as much as for hard disks. Calculate price per GB and see for yourself—but also factor in things like rpm and buffer

I recently assembled a PC for myself, and have been slowly collecting components for over three months now. Peripherals aside, my PC is complete, except from one very important component—the hard disk. Now I'm a data hog, and my backups consist of a veritable treasure trove of applications, music, games, movies, and various utilities spanning some 300-odd gigabytes!

Recent trends indicate that Serial ATA drives are completely dominating the market, and Parallel ATA is almost completely phased out. De facto spindle speeds are 7200 rpm, with 8 MB I/O buffers. In fact, many hard drives boast of SATA 2.0 features like NCQ (Native Command Queuing) and 16 MB buffers. It seems 80 GB drives are the bare minimum available, with 160 GB drives topping the sales figures. Larger capacity drives have two advantages:

1. Cost per GB is at an all-time low, and it can only get better.
2. Newer drives feature better drive electronics and the platters generally have higher areal densities, which translates to faster performance.

Hard drive warranties are also right up there, with players like Western Digital (WD) offering a five-year, at-your-doorstep replacement warranty. From a price point, 250 GB hard drives offer best bang for your buck, followed by 320 GB drives.

It's important to calculate the price per GB of a hard drive. For example, a 250 GB drive that costs Rs 3,800 gives you a price per GB of 3800/250 = 15.2, while a 160 GB drive that costs Rs 2,800 offers just 2800/160 = 17.5 Rs/GB.

The first vendor I visited at Lamington Road showed me drives from Seagate, Samsung, and WD. He had capacities ranging from 160 GB to 320 GB, and the Seagate 7200.10 series 320 GB caught my eye. This drive features a 16 MB buffer, and uses perpendicular recording. This allows for fewer platters and more storage—a good thing from a performance point of view. Also noteworthy are the Western Digital KS series, which although a touch lower on performance (as we've tested earlier) run very, very cool. Samsung's SP 2504C is another very strong performer, but most of their lower-capacity drives are outperformed by Seagate's equivalent models.

Vendor #2 had most of the same models, except he stocked Hitachi drives as well. The Hitachi T7K series features 8 MB buffers, but more importantly, they support NCQ (which older WD drives don't). In addition, Hitachi drives are known to be silent and cool. Performance is about par with WD, though the Seagate 7200.10 series drives are the fastest. This guy told me he could arrange for Raptors, which are Western Digital's performance kings for the Desktop. WD Raptors are available in 80 and 150 GB sizes, and feature 10000 rpm spindle speeds. This gives them huge performance boosts in almost every type of read/write operation. Unfortunately, the pricing (see box) will scare off nearly everybody. I also saw a couple of larger drives—a Seagate 500 GB, and a WD 400 GB. However, when accounting for price and performance, it's probably better to opt for two smaller-capacity drives rather than a single large drive.

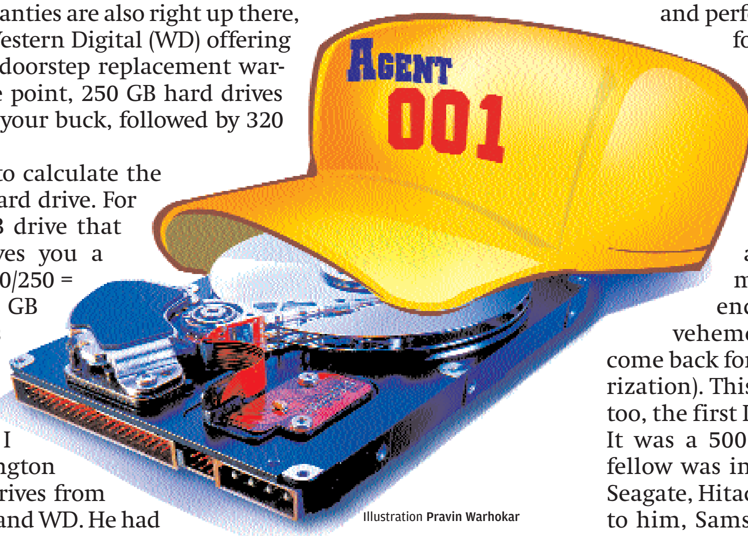

Illustration **Pravin Warhokar**

Hard Drive Prices (Rs)

Capacity	Hitachi	Samsung	Seagate	Western Digital
80 GB	2,000	1,875	2,100	2,100
160 GB	2,600	2,400	2,700	2,700
250 GB	3,500	3,480	3,650	3,650
320 GB	4,500	–	4,600	4,600
400 GB	6,450	–	6,500	6,500
500 GB	9,000	–	9,500	9,500

Vendor three showed me a similar mix of drives. Interestingly enough, all the Seagate models he had were their 7200.10 series. According to him, the 160 GB is a single platter drive, and blazingly fast. He also recommended the Seagate 320 GB. When I enquired about heating issues, he vehemently said Seagate drives rarely come back for RMA (Return Merchandise Authorization). This guy did have a Maxtor hard drive too, the first I've seen after the Seagate takeover. It was a 500 GB Diamond Max 11 drive. The fellow was insistent that I choose from one of Seagate, Hitachi or WD, in that order. According to him, Samsung and Maxtor are not the first choice when it comes to performance.

I was tempted by the storage space offered by a 400 GB drive, but the price per GB was too high. Finally, the 5-year door-to-door replacement warranty that Western Digital is offering won me over. Since speed was also important, I bought two Western Digital 2500KS drives, and plan to set up a RAID 0 configuration—which will take care of the speed parameter.

Look at the table above for the best hard drive prices I came across while looking about. ■

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